THE PROBLEM

# Lose the phone. Not the wages.

A seafarer's phone goes overboard at 03:40 off Singapore. His savings were on it. A seed-phrase wallet has no "forgot password", custodial apps want identity checks from a ship with no signal, and a company-held seed backup hands his keys to a company.

Sea-based Filipino workers sent home $7.139 billion in 2025. (BusinessWorld, citing BSP data). About 1.89 million seafarers serve the world fleet (2021 BIMCO/ICS report, via press).

**Ramon is fictional. The mechanism is real.**

WHY PORTALDOT IS ESSENTIAL

# A MetaMask address can be brought back by people it trusts.

- Every EVM address has a native twin account on Portaldot V3.
- An Ethereum transaction to the runtime pallets address runs as that twin, so MetaMask can drive the Recovery pallet.
- The rescuer runs a Solidity call as the old account: the vault sees the old address as **msg.sender** and pays out.
- Delete Portaldot and the rescue has nothing to stand on.

HOW IT WORKS

# Name a crew. Hear four words. Wait. Bring it home.

- Ramon names 3 guardians, needs 2, sets a safety delay, prints a wallet-size Lifeboat Card.
- A guardian calls him, types the four-word call phrase, starts the attempt. A second approves.
- The chain enforces the threshold and the delay. Ramon can stop any attempt he did not start.
- One signature from the new phone moves tPOT, a native asset and the vault savings, in one `force_batch`.

PROOF OF FUNCTIONALITY

# Generated from live testnet receipts.

| 2/2 | 9/9 | 0 | 0.06293 |
| --- | --- | --- | --- |
| rescues completed | attacks stopped, each decoded | honest rescues wrongly blocked | tPOT in fees per rescue |

Verify it yourself: **npm run verify** re-reads 31 recorded transactions from the public node with no key. The /proof page decodes them in your browser.

HONEST LIMITS

# What we did not prove.

- No ink! contract: the "WASM" half of WASM/EVM is not shown.
- This testnet's security deposit is 0, so a bad initiator risks only fees; the threshold and delay still hold.
- No real-wallet test, no real-people drill, no public deployment yet.
- One prediction was wrong (an error name); we report it.

REASONS FOR CONTINUED SUPPORT

# A recovery flow any Portaldot wallet can ship.

- First public measurements of the Recovery pallet on V3 (docs/FINDINGS.md).
- An open, dependency-free encoder and a protection flow other dApps can copy.
- A reachable market: seafarers and the families who depend on them, through manning agencies and welfare groups.
- Free for individuals; paid for integrations.

**github.com/0xkinno/kedge**